



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

EMM 201: COMPUTERS PROGRAMMING 1

2ND SEMESTER 2020/2021

TIME: 2 HOURS

INSTRUCTIONS:

This paper consists of **FIVE** questions

Question **ONE** is compulsory and contains 30 marks

Attempt any **TWO** of the remaining four questions. Each question is 20 marks.

Question 1 (30 marks)

(a) Define the following terms and give example for each:

(i) High level languages

(ii) Language compiler

(4 marks)

(b) Design a program to find all the roots of a quadratic equation $ax^2 + bx + c = 0$

Use flow chart

(4 marks)

(c) Write a program in C++ to enter two angles of a triangle and find the third angle.

(6 marks)

(d) Write a C++ program that computes and display volume of water to be contained in sphere of radius between 10cm to 150 cm. Program should display volume in litres.

(4 marks)

(e) Discuss any three factors to consider while selecting a programming language compiler.

(6 marks)

(f) Manufacturing Overhead is calculated using the formular given below:

Manufacturing Overhead = Cost of Goods Sold – Cost of Raw Material – Direct Labour Cost

Write a program that receive cost item form user and display manufacturing costs in kshs.

(6 marks)

Question 2(20 Marks).

- ↑ a) Using a simple diagram illustrate structure of exit point loop. [3 marks]
- ↑ b) Briefly explain stages of computer problem solving. State product of each stage. [4 marks]
- ↑ c) Differentiate between inbuilt and user defined data types. Give example in each case. [6 marks]
- * d) The following details are available for an employee: **employee Number, Name, salary**
Write a C++ program to capture details for all 25 employee details. Assume company is to effect 5 percent pay cut due to COVID-19 effects. Write a program to display old and new salaries for all employee and indicate total saving from deductions [7 marks]

Question 3(20 Marks).

- ↑ a) Outline any three characteristics of an algorithm [3 marks]
- ↑ b) Briefly outline three reasons why algorithms aren't represented using natural language. [6 marks]
- * c) Material Price Variance can be calculated using the following formula:
$$MPV = (\text{Standard Price} - \text{Actual Price}) \times \text{Actual Quantity}$$

	Standard	Actual
Price	Kshs 10 per kg.	kshs 8 per kg.

Write a program that generate MPV for quantities between 500kg and 1.2 tonnes. [6 marks]

- * d) Write a program that record each employee salary and daily meal cost at staff cafeteria for a month. Your program should display all 30 employee and total cost of meal for a month. Assume company is considering adjusting meal cost to be 0.5 percent of employee basic salary. Modify program to display previous cost and new proposed costs. [5 Marks]

Question 4(20 Marks).

- ↑ a) Briefly explain how the following product are generated
 - i) Problem requirements
 - ii) Source code
 - iii) Design[6 marks]
- ↑ b) Explain general structure of multiple selection control structure. [2 marks]
- * c) Write a program that inputs three different items' costs and outputs the cheapest of them. [4 marks]

(d) The cutting speed of a drill is expressed as

$$VC = \frac{\pi \cdot DC \cdot n}{1000} \text{ (m/min)}$$

Where

π (3.14): Circular constant

DC (mm): Drill diameter

n (min^{-1}): Spindle speed

write a program to display various VC for drill diameter between 10mm to 35 mm in steps of 3

[8 marks]

Question 5(20 Marks).

a) Define the following items

i) Code block

ii) variable scope

iii) function definition

[6 marks]

b) Briefly explain any two disadvantages of low level programming language.

[4 marks]

b) To determine the hourly fuel cost, the total fuel cost is divided by the productive time of the equipment. If fuel consumption records are not available, the following formula can be used to estimate liters of fuel used per machine hour,

$$\text{LMPH} = \frac{K \times \text{GHP} \times \text{LF}}{\text{KPL}}$$

where LMPH is the liters used per machine hour, K is the kg of fuel used per brake hp/hour, GHP is the gross engine horsepower at governed engine rpm, LF is the load factor in percent, and KPL is the weight of fuel in kg/liter.

[10 marks]

DATE: THURSDAY 17TH DECEMBER 2020

TIME: 11.00 A.M. – 1.00 P.M.

INSTRUCTIONS:

Answer Question ONE and any other TWO questions

Question 1 (Compulsory)

(a) Describe the following of terms:

- (i) Problem solving using a computer
- (ii) Constants
- (iii) Local variables
- (iv) Function definition.
- (v) Pseudocode

[10 marks]

(b) Briefly explain any two limitations of modular programming.

[2 marks]

(c) Describe the following terms as used in programming

- (i) Algorithm
- (ii) Program specification
- (iii) Problem analysis

[2 marks]

[2 marks]

[2 marks]

(d) Write a C++ program to check whether three given integer values are in the range 20..50 inclusive. Return true if 1 or more of them are in the said range otherwise return false

[6 marks]

(e) Write a function named "swap_floats" that takes two floating point arguments and interchanges the values that are stored in those arguments. The function should return no value.

[6 marks]

Question 2

(a) Write a structure to store the custno, name, gender, account balance, age (between 24 to 74) and address of customers (more than 10). capture the information of the customers.

[6 marks]

(b) Write a function to print the names of all the customers having account balance exceeding kshs 50000.

[4 marks]

(c) Write another function to print the names of all the female customers having a balance in excess of kshs 100,000.

[4 marks]

- (d) Write another function to display the details of the customer when user provide custno (i.e. custno. entered by the user).

[6 marks]

Question 3

- (a) Briefly explain the following stages of program development cycle
- i). Developing the outline into algorithm [2 marks]
 - ii). Testing and debugging [2 marks]
- (b) Write a program to add two distances in meters-centimetre using structure. The values of the distances is to be taken from the user. [6 marks]
- (c) Enter the marks of 5 students in Chemistry, Mathematics and Physics (each out of 100) using a structure named Marks having elements regno., name, chem_marks, maths_marks and phy_marks and then display the average marks of each student [10 marks]

Question 4

- (a) Outline any three qualities of a good program compiler [3 marks]
- (b) Write a function named "subtotal" takes as its arguments the following: (1) an array of floating point values; (2) an integer that tells the number of cells in the array. The function should replace the contents of each cell with the sum of the contents of all the cells in the original array from the left end to the cell in question. [6 marks]
- (c) Write a program that generates and displays both the surface area and volume of cubes of size between 3 and 25 centimetres length (assume all sides of the cube are of equal sizes) [4 marks]

Question 5

- (a) Explain main challenges of writing program using functions. [6 marks]
- (b) Write a structure to store the name, account number and balance of customers (more than 10) and store their information. (4 marks)
- (c) Write a function to print the names of all the customers having balance less than kshs 2000. (4 marks)
- (d) Write a function to add a bonus of 2.5% in the balance of all the customers having more than kshs 10000 in their balance and then print the incremented value of their balance.

(6 marks)

INSTRUCTIONS:

Answer Question One and Any Other Two questions

EXCEL CENTER KCM
Tel: 0706-146003
Date.....

QUESTION 1.

a) Differentiate the following

- i). High level languages and low level languages
- ii). Selection and repetition statements

[4 marks]

b) Describe the following program design tools

- i) Flow charts
- ii) Pseudo code

[4 marks]

c) Write a function that accepts a positive integer mark and return grades according to university grading scale.

[5 marks]

d) Write a C++ program that computes roots of quadratic equation using the mathematical formula below

$$X1 = -b + \sqrt{(a+b^2) \div 2a}$$

$$X2 = -b - \sqrt{(a+b^2) \div 2a}$$

Write a program that generates sine and cosine of numbers between 45 and 135 inclusive.

[6 marks]

Write a program to compute cost of each item below in a building given estimated cost is 25 millions shilling.

(i) Cost of planning and approvals=5%

(ii) Raw materials and transport to site=30%

(iii) Labor=35%

(iv) Supervision and other consultancy=15%

(v) Miscellaneous cost=15%

[6 marks]

QUESTION 2

a) Briefly explain process of program development. Use relevant example in your explanation

[4 marks]

b) Using appropriate example differentiate between a function and a structure

[4 marks]

c) The following details are available for an employee: employee Number, Name, salary, commission and overtime. Write a C++ program to include a structure called 'employee' to store the employee details and display the same along with total pay and income tax of 30% if total pay exceed 20,000 shillings.

The maximum number of employees should not exceed 50.

[12 marks]

QUESTION 3

a) Briefly explain the following stages of program development cycle

i). Problem definition

ii). Testing and Debugging

[4 marks]

b) Explain structure and use of array in programming.

[3 marks]

c) Write a program that inputs three different items' costs and outputs the cheapest of them.

[5 marks]

- d) Write a C++ program that allows the user to enter and store the day's hourly temperature. The program then determines and displays the lowest temperature, the highest temperature and average days' temperatures using a suitable format.

[8 marks]

Question 4.

- a) Differentiate the following set of terms:
- Code block and variable scope
 - Global variables and Local variables
 - Function prototypes and function definition

[6 marks]

- b) Briefly explain any two advantages of partitioning a program into functions

[2 marks]

- a) Write a C++ program that uses four functions namely input, Grade, Display and main to read in the material and moisture content. The input function is used to read moisture (out of 100) for a material. This function should only accept values that fall within the range 0-100. The moisture is then passed as a parameter to the Grade function which uses it to assign a grade based on the following classification:

Mark	Grade
75 and above	M1A
60 and below 75	M1B
50 and below 60	M1C
40 and below 50	M1D
Below 40	M1E

The Moisture and computed grade are the passed as parameters to the output function which outputs the appropriately. All functions input, Grade and output are called from main.

[12 marks]

- a) Outline any three characteristics of a good algorithm [3 marks]
- b) A program is required to calculate the average of a set of marks and to count the number of failing marks. The program prompts the user the number of marks and reads the value entered. The program then reads each mark and determines the failing marks as those below 40 counting each failing mark. Your program should display the sum, average and number of failing marks

Required:

- i) Draw a flowchart for your solution [7 marks]
- ii) Write the C++ program that implements the above algorithm [6 marks]

Write a program that record number and their word equivalent. The program should prompt user for a number and program display word equivalent of the same in format below

Number	Word equivalent
1	one
2	two
3	three
...	...
...	...
10	ten

[4 marks]